TRANSFORM AND BOUNDARY VALUE PROBLEMS

MATHEMATICS-III

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Introduction

Differential equations have wide applications in various engineering and science disciplines. In general, modeling of the variation of a physical quantity, such as temperature, pressure, displacement, velocity, stress, strain, current, voltage, or concentration of a pollutant, with the change of time or location, or both would result in differential equations. Similarly, studying the variation of some physical quantities on other physical quantities would also lead to differential equations.
Although almost all physical phenomena in this Universe can be represented by differential equations at microscopic as well as macroscopic, the following are very few use of it in Computer Science Engineering.

Differential Equations in Computer Science Engineering

- Lagrange multipliers are always used to optimize and maximize the functions of neural networks (AI-artificial intelligence) :
- Gradient decent methods are used in back propagation for fitting an artificial neural network:
- Almost all of machine learning and AI (svm ,regressions trees) are filled with examples of differential equations:
- In computer graphics, animation, computer vision, and shape retrieval such as denoising, fairing, feature extraction, segmentation, simplification, classification, and editing; the complex manipulations can be described precisely, compactly, and measurably by using PDEs:
- If you develop a computer programme to predict future values, especially on engineering quantities, you will almost certainly use differential equations as part of your predictive algorithms in your tool kit :

Definition: The differential equation involving more than one independent variables and any number of dependent variables is called PDE.
Let $z = f(x, y)$ then

$\frac{\partial z}{\partial x} = p =$ partial derivative of z w.r.t x

$\frac{\partial z}{\partial y} = q =$ partial derivative of z w.r.t y

$\frac{\partial^2 z}{\partial x^2} = r, \qquad \frac{\partial^2 z}{\partial y^2} = t, \qquad \frac{\partial^2 z}{\partial x \partial y} = s.$

Example: $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1, \quad \left(\frac{\partial z}{\partial x}\right)^2 + 2\left(\frac{\partial^2 z}{\partial x \partial y}\right) + \left(\frac{\partial z}{\partial y}\right)^2 = 0$ and so on....

Order: The order of the PDE is the order of the highest order derivative occurring in it.

Degree: The degree of the PDE is the power of its highest derivative occurring in it.

Example: (i) $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1,$ order=1, degree=1.

(ii) $\left(\frac{\partial z}{\partial x}\right)^2 + 2\left(\frac{\partial^2 z}{\partial x \partial y}\right) + \left(\frac{\partial z}{\partial y}\right)^2 = 0,$ order=2, degree=1.

(iii) $\left(\frac{\partial^2 z}{\partial x^2}\right)^{\frac{3}{2}} = \frac{\partial z}{\partial x} \Rightarrow \left(\frac{\partial^2 z}{\partial x^2}\right)^3 = \left(\frac{\partial z}{\partial x}\right)^2,$ order=2, degree=3.

Let

$$f(x,\ y,\ z,\ a,\ b) = 0 \tag{1}$$

be the given function, where $z = z(x, y)$ is the dependent variable depends on x and y, a and b are arbitrary constants. Then we can form a PDE from equation (1) in two ways:

- Eliminating arbitrary constants
- Eliminating arbitrary function

Let us consider equation (1) e.i. $f(x,\ y,\ z,\ a,\ b) = 0$ and we proceed as below:

Step-I: Differentiate equation (1) w.r.t x

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x} = 0 = \frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z} \tag{2}$$

Step-II: Differentiate equation (1) w.r.t y

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial y} = 0 = \frac{\partial f}{\partial x} + q\frac{\partial f}{\partial z} \tag{3}$$

Eliminate the constants a and b from equations (2) and (3).

Finally, write the PDE in the form

$$\phi(x,\ y,\ z,\ p,\ q) = 0. \tag{4}$$

Note: The equation (1) is called the primitive or complete solution of the equation (4).

Example 1: Form the PDE from $z = (x^2 + a)(y^2 + b)$.

Solution: Differentiating w.r.t x we get

$$\frac{\partial z}{\partial x} = 2x(y^2 + b) = p \tag{5}$$

Differentiating w.r.t y we get

$$\frac{\partial z}{\partial y} = 2y(x^2 + a) = q \tag{6}$$

Equation(5) and (6) gives

$$(y^2 + b) = \frac{p}{2x} \quad \text{and} \quad (x^2 + a) = \frac{q}{2y} \quad \text{respectively.} \tag{7}$$

Substituting equation (7) in the original equation
$z = (x^2 + a)(y^2 + b) = \frac{p}{2x} \cdot \frac{q}{2y} \quad \Rightarrow pq = 4zxy.$

Example 2: Find the PDE of all planes cutting equal intercepts from the x-axis and y-axis.

Solution: Let a and c are x and z intercepts respectively. The the equation of plane is given by

$$\frac{x}{a} + \frac{y}{a} + \frac{z}{c} = 1. \tag{8}$$

Differentiating w.r.t x we get

$$\frac{1}{a} + \frac{1}{c}\frac{\partial z}{\partial x} = 0 = \frac{1}{a} + \frac{1}{c}p. \tag{9}$$

Differentiating w.r.t y we get

$$\frac{1}{a} + \frac{1}{c}\frac{\partial z}{\partial y} = 0 = \frac{1}{a} + \frac{1}{c}q. \tag{10}$$

Solving (9) and (10) gives

$$\frac{1}{c}p - \frac{1}{c}q = 0 \quad \Rightarrow \frac{1}{c}(p-q) = 0 \qquad \Rightarrow p - q = 0.$$

Example 3: Form the PDE from $\log(az-1) = x + ay + b$.

Solution:

$$\frac{a}{az-1}\frac{\partial z}{\partial x} = 1 = \frac{ap}{az-1} \qquad \text{and} \tag{11}$$

$$\frac{a}{az-1}\frac{\partial z}{\partial y} = a = \frac{aq}{az-1} \tag{12}$$

Diving (12) by (11), we get $\frac{q}{p} = a$. From equation (12) we have

$$q = az - 1 = \frac{q}{p}z - 1 \quad \Rightarrow pq = qz - p \quad \Rightarrow p(q+1) = qz.$$

Example 4: Obtain the PDE of all spheres with centers lies on $z = 0$ and whose radius is constant and equal to r.

Solution: The equation of the spheres with center at $z = 0$ is written as $(x-a)^2 + (y-b)^2 + z^2 = r^2$. Differentiating this w.r.t. x and y one can get

$$2(x-a) + 2z\frac{\partial z}{\partial x} = 0 = 2(x-a) + 2zp \quad \text{and}$$
$$2(y-b) + 2z\frac{\partial z}{\partial y} = 0 = 2(y-b) + 2zq$$
$$\Rightarrow (x-a) = -zp \quad \text{and} \quad (y-b) = -zq$$
$$\Rightarrow z^2p^2 + z^2q^2 + z^2 = r^2 = z^2(p^2 + q^2 + 1).$$

Example 5: Form a PDE by eliminating a, b and c from $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

Solution:

$$\frac{2x}{a^2}+\frac{2z}{c^2}\frac{\partial z}{\partial x}=0 \qquad \text{or} \qquad c^2x+a^2z\frac{\partial z}{\partial x}=0 \tag{13}$$

$$\frac{2y}{b^2}+\frac{2z}{c^2}\frac{\partial z}{\partial y}=0 \qquad \text{or} \qquad c^2y+b^2z\frac{\partial z}{\partial y}=0 \tag{14}$$

Again differentiating (13) w.r.t y we get

$$0+a^2\left[z\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial z}{\partial x}\frac{\partial z}{\partial y}\right]=0 \quad \Rightarrow pq+zs=0. \tag{15}$$

Problems for practice.

- Find the PDE of the set of all right circular cones whose axes coincides with $z-$axis and given by $x^2 + y^2 = (z - c)^2 \tan^2 \alpha$.
  Ans: $yp = xq$
- Form a PDE by eliminating a and b, from $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
  Ans: $2z = px + qy$
- Form a PDE by eliminating a and b, from $2z = (ax + y)^2 + b$
  Ans: $q = \frac{p}{q}x + y$

In order to form a PDE by eliminating arbitrary functions, we may come across two types of functions

- **Form I:** $z = f(x, y)$, where x and y are independent variables.
- **Form II:** $\phi(u, v) = 0$, where $u = u(x, y, z)$, $v = v(x, y, z)$ and x, y and z are independent variables.

Example 1: Form the PDE by eliminating the arbitrary function from $z = f(x^2 + y^2)$.

Solution: Differentiating w.r.t x and y separately, we get

$$\frac{\partial z}{\partial x} = 2xf^{'}(x^2 + y^2) = p \quad \text{and} \quad \frac{\partial z}{\partial y} = 2yf^{'}(x^2 + y^2) = q.$$

Dividing them we get

$$\frac{p}{q} = \frac{x}{y}.$$

Example 2: Form the PDE by eliminating the arbitrary function from $z = f(x - at) + f(x + at)$.

Solution: Differentiating w.r.t x, we get

$$\begin{aligned} \frac{\partial z}{\partial x} &= f^{'}(x - at) + f^{'}(x + at) \\ \frac{\partial^2 z}{\partial x^2} &= f^{''}(x - at) + f^{''}(x + at). \end{aligned} \tag{16}$$

Similarly differentiating w.r.t t, we get

$$\begin{aligned} \frac{\partial z}{\partial t} &= -af^{'}(x - at) + af^{'}(x + at) \\ \frac{\partial^2 z}{\partial t^2} &= a^2 f^{''}(x - at) + a^2 f^{''}(x + at). \end{aligned} \tag{17}$$

Form (17) one can see that

$$\begin{aligned}\frac{\partial^2 z}{\partial t^2} &= a^2 f''(x-at) + a^2 f''(x+at) \\ &= a^2[f''(x-at) + f''(x+at)] \\ &= a^2\left[\frac{\partial^2 z}{\partial x^2}\right]. \quad \text{From (16)}\end{aligned}$$

Therefore the required PDE is $\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$.

Example 3: Form the PDE by eliminating the arbitrary function from $y = f\left(\frac{y}{x}\right)$.

Solution: Differentiating w.r.t x and y separately, we get

$$\frac{\partial z}{\partial x} = f'\left(\frac{y}{x}\right)\left(\frac{-y}{x^2}\right) \quad \text{or} \quad f'\left(\frac{y}{x}\right) = \left(\frac{-x^2}{y}\right)\frac{\partial z}{\partial x} \tag{18}$$

and

$$\frac{\partial z}{\partial y} = f'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) \quad \text{or} \quad f'\left(\frac{y}{x}\right) = x\frac{\partial z}{\partial y} \tag{19}$$

From (18) and (19) one can find

$$\left(\frac{-x^2}{y}\right)\frac{\partial z}{\partial x} = x\frac{\partial z}{\partial y} \qquad \Rightarrow x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0.$$

Example 4: Form the PDE by eliminating the arbitrary function from $y = x^2 + 2f\left(\frac{1}{y} + \log x\right)$.

Solution: Differentiating w.r.t x and y separately, we get

$$\begin{aligned}
\frac{\partial z}{\partial x} = p &= 2x + 2f'\left(\frac{1}{y} + \log x\right)\left(\frac{1}{x}\right) \quad \text{and} \\
\frac{\partial z}{\partial y} = q &= 2f'\left(\frac{1}{y} + \log x\right)\left(\frac{-1}{y^2}\right) \\
&\Rightarrow \quad px + qy^2 = 2x^2 \\
&\Rightarrow \quad \frac{p - 2x}{q} = -\frac{y^2}{x}.
\end{aligned}$$

Example 5: Form the PDE by eliminating the arbitrary function from $y = xy + 2f(x^2 + y^2 + z^2)$.

Solution: Differentiating w.r.t x and y separately, we get

$$\begin{aligned}
\frac{\partial z}{\partial x} = p &= y + 2f^{'}(x^2 + y^2 + z^2)\left(2x + 2z\frac{\partial z}{\partial x}\right) \quad \text{and} \\
\frac{\partial z}{\partial y} = q &= x + 2f^{'}(x^2 + y^2 + z^2)\left(2y + 2z\frac{\partial z}{\partial y}\right) \\
&\Rightarrow \quad p - y = 2f^{'}(x^2 + y^2 + z^2)(2x + 2zp) \\
&\Rightarrow \quad q - x = 2f^{'}(x^2 + y^2 + z^2)(2y + 2zq) \\
&\Rightarrow \quad \frac{p - y}{q - x} = \frac{x + zp}{y + zq}.
\end{aligned}$$

Problems for practice.

- Form the PDE by eliminating the arbitrary functions f and g $z = f(ax + by) + g(\alpha x + \beta y)$.

  Ans: $b\beta \frac{\partial^2 z}{\partial x^2} - (a\beta + b\alpha)\frac{\partial^z}{\partial x \partial y} + a\alpha \frac{\partial^2 z}{\partial x^2} = 0$
- Form a PDE by eliminating f and g, from $z = yf(x) + xg(y)$

  Ans: $xy\frac{\partial^z}{\partial x \partial y} = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} - z$
- Eliminate the arbitrary function f and obtain the PDE from $z = e^y f(x + y)$.

  Ans: $\frac{\partial z}{\partial y} = z + \frac{\partial z}{\partial x}$

Let $\phi(u,\ v) = 0$ be given function. Then we can construct the PDE as follows:

Step 1: Differentiate u and v w.r.t x, y and z

Step 2: Find

$$P = \frac{\partial u}{\partial y}\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\frac{\partial v}{\partial y} = \frac{\partial(u,v)}{\partial(y,z)}$$
$$Q = \frac{\partial u}{\partial z}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial v}{\partial z} = \frac{\partial(u,v)}{\partial(z,x)}$$
$$R = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} = \frac{\partial(u,v)}{\partial(x,y)}$$

Step 3: Write the PDE $Pp + Qq = R$.

Example 1: Form the PDE by eliminating the arbitrary function ϕ from $\phi(x+y+z,\ x^2+y^2-z^2)=0$.

Solution: Here $u = x+y+z$ and $v = x^2+y^2-z^2$

$$\frac{\partial u}{\partial x}=1, \quad \frac{\partial u}{\partial y}=1, \quad \frac{\partial u}{\partial z}=1$$

$$\frac{\partial v}{\partial x}=2x, \quad \frac{\partial v}{\partial y}=2y, \quad \frac{\partial v}{\partial z}=-2z.$$

Now

$$P=\frac{\partial u}{\partial y}\frac{\partial v}{\partial z}-\frac{\partial u}{\partial z}\frac{\partial v}{\partial y}=1.(-2z)-1.(2y)=-2z-2y$$

$$Q=\frac{\partial u}{\partial z}\frac{\partial v}{\partial x}-\frac{\partial u}{\partial x}\frac{\partial v}{\partial z}=1.(2x)-1.(-2z)=2x+2z$$

$$R=\frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial u}{\partial y}\frac{\partial v}{\partial x}=1.(2y)-1.(2x)=2y-2x.$$

Write the PDE as

$$
\begin{aligned}
& Pp + Qq = R \\
& \Rightarrow (-2z - 2y)p + (2x + 2z)q = 2y - 2x \\
& = (y + z)p - (x + z)q = x - y
\end{aligned}
$$

Example 2: Form the PDE by eliminating the arbitrary function ϕ from $\phi(x^2+y^2+z^2,\ z^2-2xy)=0$.

Solution: Here $u=x^2+y^2+z^2$ and $v=z^2-2xy$

$$\frac{\partial u}{\partial x}=2x,\quad \frac{\partial u}{\partial y}=2y,\quad \frac{\partial u}{\partial z}=2z$$
$$\frac{\partial v}{\partial x}=-2y,\quad \frac{\partial v}{\partial y}=-2x,\quad \frac{\partial v}{\partial z}=2z.$$

$$\begin{aligned} P &= (2y)(2z)-(2z)(-2x)=4z(y+x)\\ Q &= (2z)(-2y)-(2x)(2z)=-4z(y+x)\\ R &= (2x)(-2x)-(2y)(-2y)=4(y^2-x^2). \end{aligned}$$

Hence $Pp+Qq=R \Rightarrow p-q=\frac{y-x}{z}$.

Example 3: Form the PDE by eliminating the arbitrary function f from $f\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right) = 0$.

Solution: Here $u = \frac{x-a}{z-c}$ and $v = \frac{y-b}{z-c}$

$$\frac{\partial u}{\partial x} = \frac{1}{z-c}, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial u}{\partial z} = -\frac{x-a}{(z-c)^2}$$
$$\frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = \frac{1}{z-c}, \quad \frac{\partial v}{\partial z} = -\frac{y-b}{(z-c)^2}.$$

$$\begin{aligned}
\Rightarrow P &= (0)\left(-\frac{y-b}{(z-c)^2}\right) - \left(-\frac{x-a}{(z-c)^2}\right)\frac{1}{z-c} = \frac{x-a}{(z-c)^3} \\
Q &= \left(-\frac{x-a}{(z-c)^2}\right)(0) - \frac{1}{z-c}\left(-\frac{y-b}{(z-c)^2}\right) = \frac{y-b}{(z-c)^3} \\
R &= \frac{1}{z-c}.\frac{1}{z-c} - (0)(0) = \frac{1}{(z-c)^2}.
\end{aligned}$$

Hence the PDE is

$$\begin{aligned}
Pp + Qq &= R \\
(x-a)p + (y-b)q &= (z-c).
\end{aligned}$$

Example 4: Form the PDE by eliminating the arbitrary function ϕ from the relation $\phi(x^2+y^2+z^2,\ lx+my+nz)=0$.

Solution: Here $u=x^2+y^2+z^2$ and $v=lx+my+nz$

$$\begin{aligned} P &= \frac{\partial u}{\partial y}\frac{\partial v}{\partial z}-\frac{\partial u}{\partial z}\frac{\partial v}{\partial y}=(2y).n-(2z).m \\ Q &= \frac{\partial u}{\partial z}\frac{\partial v}{\partial x}-\frac{\partial u}{\partial x}\frac{\partial v}{\partial z}=(2z).l-(2x).m \\ R &= \frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial u}{\partial y}\frac{\partial v}{\partial x}=(2x).m-(2y).n. \end{aligned}$$

$$Pp+Qq=R \Rightarrow 2(ny-mz)p+2(lz-nx)q=2(mx-ly)$$

$$=(ny-mz)p+(lz-nx)q=(mx-ly).$$

Solution (Integral): The solution of a PDE is a relation between the independent variables and dependent variables which satisfies the given PDE.

There are four types of solutions:

- **Complete Integral**
- **Particular Integral**
- **Singular Integral**
- **General Integral**.

Definition: A solution which contains as many arbitrary constants as there are independent variables is complete integral.

Example: If $f(x, y, z, p, q) = 0$ where x and y are independent variables is a given PDE, then the solution of the form $\phi(x, y, z, a, b) = 0$ is a complete integral.

Definition: Let $\phi(x, y, z, a, b) = 0$ be a complete integral of the given PDE where a and b are arbitrary constants. If we give particular values to a and b, we get particular integral.

Example: If $\phi(x, y, z, a, b) = 0$ is a complete integral, then $\phi(x, y, z, 1, 2) = 0$, $\phi(x, y, z, 0, 1) = 0$, $\phi(x, y, z, 3, 9) = 0$ all are possible complete integral.

Definition: Let

$$\phi(x, y, z, a, b) = 0 \tag{20}$$

be a complete integral of a given PDE. Differentiating (20) w.r.t a and b, one can obtain

$$\frac{\partial \phi}{\partial a} = 0 \qquad \text{and} \qquad \frac{\partial \phi}{\partial b} = 0. \tag{21}$$

Now eliminating a and b from (20) and (21) we get singular integral.

Definition: Let $\phi(x, y, z, a, b) = 0$ be a complete integral of $f(x, y, z, p, q) = 0$. If we take $b = \varphi(a)$ or $a = \varphi(b)$ we get

$$\phi(x, y, z, a, \varphi(a)) = 0 \qquad \text{or} \qquad \phi(x, y, z, \varphi(b), b) = 0 \tag{22}$$

Differentiating (22) w.r.t a we can obtain

$$\frac{\partial \phi}{\partial a} = 0. \tag{23}$$

Now eliminating a from (22) and (23) we get general integral.

In order to find the solutions of some first order non-linear PDEs, let us classify the given PDE as below:

- **Type I**
- **Type II (Clairaut's form)**
- **Type III**
- **Type IV (Separable equations)**.

Let us consider the first order PDEs of the form $f(x,\ y,\ z,\ p,\ q) = 0$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$

Type I: If $F(p,\ q) = 0$, i.e it contains only p and q then the solution can be obtained as below:

- **Step 1 :** Put $p = a$ and $q = b$ which gives $f(a,\ b) = 0$
- **Step 2 :** Assume the solution as $z = ax + by + c$
- **Step 3 :** From $f(a,\ b) = 0$, write $a = \varphi(b)$ or $b = \varphi(a)$
- **Step 4 (Complete Integral):** Write the complete integral as

$$z = ax + \varphi(a)y + c \tag{24}$$

- **Step 5 (Singular solution):** Obtain the singular solution by eliminating a and c from the equations (24) and

$$\frac{\partial z}{\partial a} = 0, \qquad \frac{\partial z}{\partial b} = 0$$

- **Step 6 (General solution):** Assume $c = \psi(a)$ and eliminate a from the following equations

$$z = ax + \varphi(a)y + \psi(a) \qquad \text{and} \qquad \frac{\partial z}{\partial a} = 0 = x + \varphi^{'}(a)y + \psi^{'}(a).$$

Example 1: Solve $\sqrt{p}+\sqrt{q}=1$.

Solution: Consider

$$F(p,q)=\sqrt{p}+\sqrt{q}-1 \Rightarrow F(a,b)=\sqrt{a}+\sqrt{b}-1.$$

Write the complete integral as

$$\begin{aligned} z &= ax+by+c \qquad \text{with} \qquad \sqrt{a}+\sqrt{b}=1. \\ &\Rightarrow \sqrt{b}=1-\sqrt{a} \qquad \Rightarrow b=(1-\sqrt{a})^2 \end{aligned}$$

Therefore, the complete integral is given by

$$z = ax+(1-\sqrt{a})^2y+c, \tag{25}$$

where a and c are arbitrary constants.

For the singular integral, differentiate (25) w.r.t c and equate with 0, we get

$$\frac{\partial z}{\partial c} = 0, \qquad \Rightarrow 1 = 0, \quad \text{which is absurd.}$$

Therefore, there is no singular solution. For the general solution take $c = \psi(a)$, hence we get

$$z \;=\; ax + (1-\sqrt{a})^2 y + \psi(a). \tag{26}$$

Now differentiation of (26) w.r.t a gives

$$\frac{\partial z}{\partial a} = x + 2(1-\sqrt{a}) \times \frac{-1}{2\sqrt{a}} + \psi^{'}(a) = 0. \tag{27}$$

Eliminating a from (26) and (27), we get general solution.

Example 1: Solve $p^2 + q^2 = npq$.

Solution: Consider

$$\begin{aligned} F(p,q) &= p^2 + q^2 - npq = 0 \qquad \Rightarrow F(a,b) = a^2 + b^2 - nab = 0 \\ &\Rightarrow b = \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right]. \end{aligned}$$

Complete integral:

$$z = ax + \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right] y + c.$$

Singular integral: Differentiating the above equation w.r.t c

$$\frac{\partial z}{\partial c} = 0, \qquad \Rightarrow 1 = 0, \quad \text{which is absurd.}$$

Therefore, there is no singular solution.

For the general solution take $c = \psi(a)$, hence we get

$$z = ax + \frac{a}{2}\left[n \pm \sqrt{n^2 - 4}\right] y + \psi(a). \tag{28}$$

Now differentiation of (28) w.r.t a gives

$$\frac{\partial z}{\partial a} = x + \frac{1}{2} \times \left[n \pm \sqrt{n^2 - 4}\right] y + \psi^{'}(a) = 0. \tag{29}$$

Eliminating a from (28) and (29), we get general solution.

Type II(Clairaut's form): Unit-I

Let the first order PDEs is of the form $z = px + qy + f(p, q)$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. then the solution can be obtained as below:

- **Complete integral:** Put $p = a$ and $q = b$ and write the complete integral

$$z = ax + by + f(a, b). \tag{30}$$

- **Singular integral:** Obtain the singular integral by eliminating a and b from (30) and the following equations

$$\text{Diff. w.t.t } a \quad \Rightarrow x + \frac{\partial f}{\partial a} = 0 \quad \text{and} \tag{31}$$

$$\text{Diff. w.t.t } a \quad \Rightarrow y + \frac{\partial f}{\partial b} = 0. \tag{32}$$

- **General integral:** Putting $b = \psi(a)$ in (30), we get

$$z = ax + \psi(a)y + f(a, \psi(a)). \tag{33}$$

Differentiation of (33) w.r.t a gives

$$\frac{\partial z}{\partial a} = x + \psi^{'}(a)y + f^{'}(a) = 0. \tag{34}$$

Eliminating a from (33) and (34), one can get the general integral.

Example 1: Solve $z = px + qy + \sqrt{1 + p^2 + q^2}$.

Solution: This is of the form $z = px + qy + f(p, q)$ where $f(p, q) = \sqrt{1 + p^2 + q^2}$.
Complete integral: It is given by $z = ax + by + \sqrt{1 + a^2 + b^2}$.

$$z = ax + by + \sqrt{1 + a^2 + b^2}. \tag{35}$$

Singular integral: Differentiate (35) w.r.t a and b and eliminate them as follows:

$$x + \frac{a}{\sqrt{1 + a^2 + b^2}} = 0 \Rightarrow x = -\frac{a}{\sqrt{1 + a^2 + b^2}}$$

and

$$y + \frac{b}{\sqrt{1 + a^2 + b^2}} = 0 \Rightarrow y = -\frac{b}{\sqrt{1 + a^2 + b^2}}. \tag{36}$$

From the above equations one can find

$$\begin{aligned} x^2+y^2 &= \frac{a^2}{1+a^2+b^2}+\frac{b^2}{1+a^2+b^2} \\ \Rightarrow 1-x^2-y^2 &= \frac{1}{1+a^2+b^2} \\ \Rightarrow 1+a^2+b^2 &= \frac{1}{1-x^2-y^2} \quad \Rightarrow \sqrt{1+a^2+b^2}=\frac{1}{\sqrt{1-x^2-y^2}}. \end{aligned}$$

As a result (36) becomes

$$\begin{aligned} x &= -a\sqrt{1-x^2-y^2} \qquad \text{and} \qquad y=-b\sqrt{1-x^2-y^2} \\ \Rightarrow a &= -\frac{x}{\sqrt{1-x^2-y^2}} \qquad \text{and} \qquad b=-\frac{y}{\sqrt{1-x^2-y^2}}. \end{aligned}$$

Therefore the singular integral w.r.t the equation (35) is given by

$$\begin{aligned} z &= -\frac{x^2}{\sqrt{1-x^2-y^2}} - \frac{y^2}{\sqrt{1-x^2-y^2}} + \frac{1}{1-x^2-y^2} \\ z &= \frac{1-x^2-y^2}{\sqrt{1-x^2-y^2}} = \sqrt{1-x^2-y^2} \\ &\Rightarrow \quad z^2 = 1-x^2-y^2 \end{aligned}$$

$$\boxed{z^2+x^2+y^2=1}.$$

Example 2: Solve $z = px + qy + p^2q^2$.

Solution: This is of the form $z = px + qy + f(p, q)$ where $f(p, q) = p^2q^2$.
Complete integral: It is given by

$$z = ax + by + a^2b^2. \tag{37}$$

Singular integral: Differentiate (37) w.r.t a and b and eliminate them as follows:

$$\begin{aligned} x + 2ab^2 &= 0 \quad \Rightarrow x = -2ab^2 \\ &\text{and} \\ y + 2a^2b &= 0 \quad \Rightarrow y = -2a^2b. \end{aligned} \tag{38}$$

From the above equations one can find

$$\frac{x}{a} = \frac{y}{b} = -2ab = \frac{1}{k} \quad \Rightarrow a = kx \qquad \text{and} \qquad b = ky. \tag{39}$$

Using (39) in (39) we get

$$\begin{aligned} x &= -2k^3x^2y \quad \Rightarrow k^3 = -\frac{1}{2xy} \\ z &= kxy + kxy + k^4x^2y^2 = 2kxy + kx^2y^2\left(-\frac{1}{2xy}\right) = \frac{3}{2}kxy \\ \Rightarrow z^3 &= \frac{27}{8}k^3x^3y^3 = \frac{27}{8}k^3x^3y^3\left(-\frac{1}{2xy}\right) = -\frac{27}{16}x^2y^2 \end{aligned}$$

$$\boxed{16z^3 + 27x^2y^2 = 0}.$$

Example 3: Solve $z = px + qy + p^2 - q^2$.

Solution: This is of the form $z = px + qy + f(p, q)$ where $f(p, q) = p^2 - q^2$.
Complete integral: It is given by

$$z = ax + by + a^2 - b^2. \tag{40}$$

Singular integral: Differentiate (40) w.r.t a and b and eliminate them as follows:

$$x + 2a = 0 \quad \Rightarrow x = -2a \qquad \text{and} \qquad y - 2b = 0 \quad \Rightarrow y = 2b.$$
$$\Rightarrow a = -\frac{x}{2} \qquad \text{and} \qquad b = -\frac{y}{2} \quad \Rightarrow z = -\frac{x^2}{2} + \frac{y^2}{2} + \frac{x^2 - y^2}{4}$$

$$\boxed{4z = y^2 - x^2}.$$

General integral: Put $b = \psi(a)$ in (40) becomes

$$z = ax + \psi(a)y + a^2 - (\psi(a))^2.$$

Differentiating the above equation w.r.t a we find

$$0 = x + \psi^{'}(a)y + 2a - 2\psi(a)\psi^{'}(a).$$

Eliminating a from the above equations, we will get the general solution.

The PDEs of the form $F(z, p, q) = 0$ which does not contain x and y is called Type-III problems. Assume the trial solution $u = x + ay$ and take

$$\begin{aligned} z &= f(x + by) = f(u) \\ \Rightarrow p &= \frac{\partial z}{\partial x} = \frac{dz}{du}.\frac{\partial u}{\partial x} = \frac{dz}{du} \quad \text{and} \quad q = \frac{\partial z}{\partial y} = \frac{dz}{du}.\frac{\partial u}{\partial y} = a\frac{dz}{du}. \end{aligned}$$

Then find the solutions as follows:

- **Step I:** Putting $p = \frac{dz}{du}$ and $q = a\frac{dz}{du}$ with $u = x + by$, we will get the ODE of the form $\frac{dz}{du} = \phi(z, a)$.
- **Step II:** Solve the ODE $\frac{dz}{du} = \phi(z, a)$ as $\frac{dz}{\phi(z,a)} = du$. Integrating this we get $\int \frac{dz}{\phi(z,a)} = u + c = x + by + c$, where c is arbitrary constant.
- **Step III:** Obtain the singular and general integral as usual.

Example 1: Solve $p(1+q) = qz$.

Solution: This is of the form Type III. Assume $u = x + by$ and put $p = \frac{dz}{du}$ and $q = a\frac{dz}{du}$ in the given equation.

$$\frac{dz}{du}\left(1 + a\frac{dz}{du}\right) = az\frac{dz}{du} \quad \Rightarrow a\left(\frac{dz}{du}\right)^2 + (1-az)\frac{dz}{du} = 0$$

$$\Rightarrow a\frac{dz}{du} = (1-az) \quad \Rightarrow a\frac{dz}{(1-az)} = du$$

$$\Rightarrow a\int \frac{dz}{(1-az)} = \int du + c \quad \Rightarrow \log(1-az) = x + ay + c.$$

Therefore, the complete integral is

$$\boxed{\log(1-az) = x + ay + c}.$$

Singular integral: Differentiate w.r.t a and c we get

$$\frac{z}{(1-az)} = y \qquad\qquad 0 = 1, \qquad \text{which is absurd.}$$

$\therefore$ Singular solution does not exist.
General integral: Put $c = \psi(a)$ in $\log(1-az) = x + ay + c$ differentiate it w.r.t a

$$\frac{z}{(1-az)} = y + \psi^{'}(a)$$

and eliminating $a$from the above we will get the general integral.

Example 2: Solve $9(p^2z + q^2) = 4$.

Solution: This is of the form Type III. Assume $u = x + ay$ and put $p = \frac{dz}{du}$ and $q = a\frac{dz}{du}$ in the given equation.

$$9\left[z\left(\frac{dz}{du}\right)^2 + a^2\left(\frac{dz}{du}\right)^2\right] = 4 \quad \Rightarrow 9\left(\frac{dz}{du}\right)^2(z + a^2) = 4$$

$$\Rightarrow \left(\frac{dz}{du}\right)^2 = \frac{4}{9(z + a^2)} \quad \Rightarrow \frac{dz}{du} = \frac{2}{3\sqrt{(z + a^2)}}$$

$$\Rightarrow 3\sqrt{(z + a^2)}dz = 2du \quad \text{integrating we get} \quad (z + a^2)^{3/2} = u + b$$

$\therefore$ The complete integral is given by

$$\boxed{(z + a^2)^3 = (x + ay + b)^2}$$

Suppose the equation is of the form $f(x, p, q) = 0$, which contains x, p and q. This equation is also Type III. To solve such equation we use the following steps:

- **Step I:** Assume $q = a$ and write $p = \phi(x, a)$ from $f(x, p, a) = 0$
- **Step II:** Put the values of p and q in $dz = pdx + qdy$ which gives $dz = \phi(x, a)dx + qdy$
- **Step III:** Integrate the above equation and write the complete integral as $z = \phi_1(x, a) + ay + c$, where c is integrating constant and $\phi_1(x, a)$ is integration of $\phi(x, a)$ w.r.t x.

Example 1: Solve $q = px + p^2$.

Solution: This is of the form Type III.

$$p^2 + px - q = 0, \quad \text{put } q = a \quad \Rightarrow p^2 + px - a = 0.$$

The above equation is a quadratic equation in term of p. Therefore, $p = \frac{-x \pm \sqrt{x^2+4a}}{2} = \phi(x, a)$.

$$\begin{aligned} dz &= pdx + qdy \\ &= \frac{-x \pm \sqrt{x^2+4a}}{2} dx + ady = \left[\frac{-x}{2} \pm \frac{\sqrt{x^2+4a}}{2}\right] dx + ady. \end{aligned}$$

Integrating this, we get

$$\boxed{z = \frac{-x^2}{4} \pm \frac{1}{2}\left[\sin h^{-1}\left(\frac{x}{2\sqrt{a}}\right) + \frac{x}{2}\sqrt{x^2+4a}\right] + ay + b}$$

Suppose the equation is of the form $f(y, p, q) = 0$, which contains y, p and q. This equation is also Type III. To solve such equation we use the following steps:

- **Step I:** Assume $p = a$ and write $q = \phi(y, a)$ from $f(y, p, a) = 0$
- **Step II:** Put the values of p and q in $dz = pdx + qdy$ which gives $dz = adx + \phi(y, a)dy$
- **Step III:** Integrate the above equation and write the complete integral as $z = ax + \phi_1(y, a)y + c$, where c is integrating constant and $\phi_1(y, a)$ is integration of $\phi(y, a)$ w.r.t y.

Example 1: Solve $pq = y$.

Solution: Put $p = a \quad \Rightarrow \quad q = \frac{y}{a}$. Now

$$dz \quad = \quad pdx + qdy = adx + \frac{y}{a}dy \qquad \Rightarrow adz = a^2dx + ydy.$$

Integrating this, we get

$$\boxed{az = ax^2 + y^2/2 + c}$$

is the complete integral. There will be no singular integral and the general integral can be obtain by taking $b = \psi(a)$ and eliminating a.

The first order PDE is said to be separable equation it it can put in the form $f(x,p) = \phi(y,p)$.

For such PDE the solutions can be obtained as:

- **Step I:** Put $f(x,p) = \phi(y,p) = a$
- **Step II:** Write $p = f_1(x,a)$ and $q = \phi_1(y,a)$
- **Step III:** Putting in $dz = pdx + qdy$ and integrating, get the complete integral as

$$z = \int f_1(x,a)dx + \int \phi_1(y,a)dy + b.$$

Example 1: Solve $p^2y(1+x^2) = qx^2$.

Solution: This is separable equation because

$$\frac{p^2(1+x^2)}{x^2} = \frac{q}{y} = a \quad \Rightarrow p^2 = \frac{ax^2}{(1+x^2)} \Rightarrow p = \frac{\sqrt{a}x}{\sqrt{1+x^2}} \quad \text{and} \quad q = ay.$$

$$dz \quad = \quad pdx + qdy = \frac{\sqrt{a}x}{\sqrt{1+x^2}}dx + aydy.$$

Integrating the above equation we get the complete integral as

$$\boxed{z = \sqrt{a(1+x^2)} + \tfrac{ay^2}{2} + b}$$

Example 2: Solve $p^2 + q^2 = x + y$.

Solution: This is separable equation because

$$p^2 - x = y - q^2 = a \quad \Rightarrow \quad p^2 = x + a \Rightarrow p = \sqrt{x+a} \quad \text{and}$$
$$q^2 = y - a \Rightarrow q = \sqrt{y-a}.$$

Now

$$dz = pdx + qdy = \sqrt{x+a}dx + \sqrt{y-a}dy.$$

Integrating the above equation we get the complete integral as

$$\boxed{z = \tfrac{2}{3}(x+a)^{3/2} + \tfrac{2}{3}(y-a)^{3/2} + b}$$

Example 3: Solve $p^2 + q^2 = x^2 + y^2$.

Solution: This is separable equation because

$$p^2 - x^2 = y^2 - q^2 = a^2 \quad \Rightarrow \quad p^2 = x^2 + a^2 \Rightarrow p = \sqrt{x^2 + a^2} \quad \text{and}$$
$$q^2 = y^2 - a^2 \Rightarrow q = \sqrt{y^2 - a^2}.$$

Now

$$dz = pdx + qdy = \sqrt{x^2 + a^2}dx + \sqrt{y^2 - a^2}dy.$$

Integrating the above equation we get the complete integral as

$$\boxed{z = \frac{a^2}{2}\sin h^{-1}\frac{x}{a} + \frac{x}{2}\sqrt{x^2 + a^2} + \frac{y}{2}\sqrt{y^2 - a^2} - \frac{a^2}{2}\cos h^{-1}\frac{y}{a} + b}$$

A linear first order PDE of the form

$$Pp + Qq = R, \tag{41}$$

where P, Q, R are are functions of x, y, z is called the Lagrange's linear equation. This equation can be solved in two ways

- **Method I (Direct method):**
- **Method II (Method of multipliers):**

The following steps are followed to solve (41):

- **Step 1:** Form the auxiliary simultaneous equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$
- **Step 2:** Solve the auxiliary equation to get two independent solutions $u(x, y, z) = a$ and $v(x, y, z) = b$
- **Step 3:** Write the solution as $\phi(u, v) = 0$, where ϕ is an arbitrary function.

Example 1: Find the general solution of $px + qy = z$.

Solution: Write the auxiliary equation as

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

Consider the following pair of equations and solve them

$$\frac{dx}{x} = \frac{dy}{y} \quad \text{and} \quad \frac{dy}{y} = \frac{dz}{z}$$

$$\Rightarrow \quad \frac{x}{y} = u \quad \text{and} \quad \frac{y}{z} = v.$$

$\therefore$ The general solution is given by

$$\boxed{\phi\left(\tfrac{x}{y}, \tfrac{y}{z}\right) = 0}$$

Example 2: Solve $\frac{y^2z}{x}p + xzq = y^2$.

Solution: Write the auxiliary equation as

$$\frac{xdx}{y^2z} = \frac{dy}{xz} = \frac{dz}{y^2}$$

Consider the following pair of equations and solve them

$$\frac{xdx}{y^2z} = \frac{dy}{xz} \quad \Rightarrow x^2dx = y^2dy \quad \Rightarrow x^3 - y^3 = u$$
$$\frac{xdx}{y^2z} = \frac{dz}{y^2} \quad \Rightarrow xdx = ydy \quad \Rightarrow x^2 - y^2 = v$$

$\therefore$ The general solution is given by

$$\boxed{\phi(x^3 - y^3, x^2 - y^2) = 0}$$

Example 3: Solve $\tan xp + \tan yq = \tan z$.

Solution: Write the auxiliary equation as

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z} \quad \Rightarrow \cot xdx = \cot ydy = \cot zdz$$

Taking $\cot xdx = \cot ydy$ and integrating one can get

$$\log sinx = \log siny + \log u \;\Rightarrow \frac{sinx}{siny} = u.$$

Similarly taking

$$\cot ydy = \cot zdz \quad \text{and integrating we get} \quad \frac{siny}{sinz} = v.$$

$\therefore$ The general solution is given by

$$\boxed{\phi\left(\frac{sinx}{siny}, \frac{siny}{sinz}\right) = 0}$$

The following steps are followed to solve (41):

- **Step 1:** Write the auxiliary equations as $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$
- **Step 2:** Chose the multipliers $l,\ m,\ n$ and $l^{'},\ m^{'},\ n^{'}$ and rewrite the auxiliary equations as $\frac{ldx+mdy+ndz}{lP+mQ+nR} = \frac{l^{'}dx+m^{'}dy+n^{'}dz}{l^{'}P+m^{'}Q+n^{'}R}$ such that $lP + mQ + nR = 0$ and $l^{'}P + m^{'}Q + n^{'}R = 0$
- **Step 3:** Then solve $ldx + mdy + ndz = 0$ and write the solution as $u = u(x, y, z)$. Similarly solve $l^{'}dx + m^{'}dy + n^{'}dz = 0$ and write the solution as $v = v(x, y, z)$.
- **Step 4:** Write the general solution as $\phi(u, v) = 0$, where ϕ is an arbitrary function.

Example 1: Solve $(mz - ny)p + (nx - lz)q = (ly - mx)$.

Solution: Write the auxiliary equation as

$$\frac{dx}{(mz - ny)} = \frac{dy}{(nx - lz)} = \frac{dz}{(ly - mx)}$$

Consider the multipliers as $x,\ y,\ z$

$$\frac{xdx + ydy + zdz}{x(mz - ny) + y(nx - lz) + z(ly - mx)} = \frac{xdx + ydy + zdz}{0}.$$

Solving the following

$$xdx + ydy + zdz = 0 \quad \text{we get} \quad u(x, y, z) = x^2 + y^2 + z^2.$$

Similarly considering the multipliers as $l,\ m,\ n$ we have

$$\frac{ldx + mdy + ndz}{l(mz - ny) + m(nx - lz) + n(ly - mx)} = \frac{ldx + mdy + ndz}{0}$$

Solving the following

$$ldx + mdy + ndz = 0 \quad \text{we get} \quad v(x, y, z) = lx + my + nz.$$

$\therefore$ The general solution is given by

$$\boxed{\phi(x^2 + y^2 + z^2, lx + my + nz) = 0}$$

Example 2: Solve $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$.

Solution: Write the auxiliary equation as

$$\frac{dx}{x(z^2 - y^2)} = \frac{dy}{y(x^2 - z^2)} = \frac{dz}{z(y^2 - x^2)}$$

Consider the multipliers as $x,\ y,\ z$

$$\frac{xdx + ydy + zdz}{x^2(z^2 - y^2) + y^2(x^2 - z^2) + z^2(y^2 - x^2)} = \frac{xdx + ydy + zdz}{0}.$$

Solving the following

$$xdx + ydy + zdz = 0 \quad \text{we get} \quad u(x, y, z) = x^2 + y^2 + z^2.$$

Similarly considering the multipliers as $1/x,\ 1/y,\ 1/z$ we have

$$\frac{1/xdx + 1/ydy + 1/zdz}{(z^2 - y^2) + (x^2 - z^2) + (y^2 - x^2)} = \frac{1/xdx + 1/ydy + 1/zdz}{0}.$$

Solving the following

$$1/xdx + 1/ydy + 1/zdz = 0 \quad \text{we get} \quad v(x, y, z) = xyz.$$

$\therefore$ The general solution is given by

$$\boxed{\phi(x^2 + y^2 + z^2, xyz) = 0}$$

Example 3: Solve $(y+z)p+(z+x)q=(x+y)$.

Solution: Write the auxiliary equation as

$$\frac{dx}{(y+z)}=\frac{dy}{(z+x)}=\frac{dz}{(x+y)}$$

It can also be written as below:

$$\frac{dx+dy+dz}{2(x+y+z)}=\frac{dx-dy}{y-x}=\frac{dy-dz}{z-y}.$$ Now taking

$$\frac{dx+dy+dz}{2(x+y+z)}=\frac{dx-dy}{-(x-y)}$$ and integrating we get

$$\log(x+y+z)=-2\log(x-y)+\log u \quad \Rightarrow u=(x+y+z)(x-y)^2$$

Similarly considering

$$\frac{dx-dy}{y-x}=\frac{dy-dz}{z-y} \quad \Rightarrow \frac{d(x-y)}{x-y}=\frac{d(y-z)}{y-z}$$

and integrating we get

$$\log(x-y)=\log(y-z)+\log v \quad \Rightarrow v=\frac{(x-y)}{(y-z)}$$

$\therefore$ The general solution is given by

$$\boxed{\phi\left((x+y+z)(x-y)^2, \tfrac{(x-y)}{(y-z)}\right)=0}$$

Example 4: Solve $zp + yq = x$.

Solution: Write the auxiliary equation as

$$\frac{dx}{z} = \frac{dy}{y} = \frac{dz}{x}$$

Considering the following

$$\frac{dx}{z} = \frac{dz}{x} \quad \text{and integrating we get}$$

$$u = x^2 - z^2$$

Similarly considering

$$\frac{dx+dy+dz}{x+y+z}=\frac{dy}{y} \quad \Rightarrow \frac{d(x+y+z)}{x+y+z}=\frac{dy}{y}$$

and integrating we get

$$\log(x+y+z)=\log y+\log v \quad \Rightarrow v=\frac{(x+y+z)}{y}$$

$\therefore$ The general solution is given by

$$\boxed{\phi\left(x^2-z^2, \frac{(x+y+z)}{y}\right)=0}$$

Although PDEs of Higher Order can be classify in many categories but in our course we will discuss only the following types of PDE:

- **Homogeneous Linear PDE:**
- **Non-homogeneous Linear PDE:**

The PDE of the form

$$a_0\frac{\partial^n z}{\partial x^n}+a_1\frac{\partial^n z}{\partial x^{n-1}\partial y}+a_2\frac{\partial^n z}{\partial x^{n-2}\partial y^2}+.....++a_n\frac{\partial^n z}{\partial y^n}=F(x,\ y),$$

where $a_0,\ a_1,\ .....\ a_n$ are the constants and $F(x,\ y)$ is the known functions of x and y is known as homogeneous linear PDE with constants co-efficient.
If we take $\frac{\partial}{\partial x}=D$ and $\frac{\partial}{\partial y}=D'$ in the above equation, then it can be written as

$$\left(a_0D^n+a_1D^{n-1}D'+a_2D^{n-2}D'^2+.....+a_nD'^n\right)z=F(x,\ y)$$

which can be further written as

$$f\left(D,\ D'\right)z=F(x,\ y). \tag{42}$$

The solution of (42) consist of two parts:

- **Complementary function:**
- **Particular Integral:**

In order to find the complementary function the following steps may be followed:

- **Step 1:** Consider $f\left(D,\ D'\right)z = 0$
- **Step 2:** Put $D = m$ and $D' = 1$ and write the auxiliary equation as

$$a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + ... + a_n = 0. \tag{43}$$

- **Step 3:** Solve the auxiliary equation (43) and find the roots. Let $m_1,\ m_2,\ .....m_n$ are the n roots of this equation, then write the complementary solution as per the following cases:

Case-I: If $m_1 \neq m_2 \neq ..... \neq m_n$ i.e. that is all the roots are distinct, then write the complementary function as:

$$\boxed{z = \phi_1(y+m_1x)+\phi_2(y+m_2x)+\phi_3(y+m_3x)+.....+\phi_n(y+m_nx).}$$

Case II: If $m_1 = m_2 = ... = m_n$ i.e. all (may be some) roots are equal (repeated) then write the complementary function as:

$$\boxed{z = \phi_1(y+m_1x)+x\phi_2(y+m_1x)+x^2\phi_3(y+m_1x)+....+x^{n-1}\phi_n(y+m_1x).}$$

Note: The degree of x may varies depending up on the repetition of the particular root.

Here we will discuss 6 possible types of functions to construct the P.I:

Type 1: If $F(x,\ y) = e^{ax+by}$, then P.I is given by

$$\boxed{\text{P.I} = \frac{1}{f(D,\ D')} e^{ax+by} = \frac{1}{f(a,\ b)} e^{ax+by}, \qquad f\,(a,\ b) \neq 0}$$

Note: Suppose $f\,(a,\ b) = 0$, then multiply the numerator by x and differentiate the denominator by D, as a result we get

$$\boxed{\frac{1}{f(D,\ D')} e^{ax+by} = \frac{x^n}{n!} e^{ax+by}.}$$

Type 2: If $F(x,\ y) = \sin(ax+by)$ or $\cos(ax+by)$, then P.I is given by

$$\begin{aligned} \text{P.I} &= \frac{1}{f\left(D^2,\ DD',\ D'^2\right)} \sin(ax+by)/\cos(ax+by) \\ &= \frac{1}{f\left(-a^2,\ -ab,\ -b^2\right)} \sin(ax+by)/\cos(ax+by), \end{aligned}$$

provided $f\left(-a^2,\ -ab,\ -b^2\right) \neq 0$.

Note: Suppose $f\left(-a^2,\ -ab,\ -b^2\right) = 0$, then follow the previous note.

Type 3: If $F(x,\ y) = x^r y^s$, then P.I is given by

$$\boxed{\text{P.I} = \frac{1}{f(D,\ D')} x^r y^s,}$$

where we can expand $[f\,(D,\ D')]^{-1}$ using binomial expansion in power of D and D'.

Type 4: If $F(x,\ y) = \phi(x,y)e^{ax+by}$, then P.I is given by

$$\text{P.I} \quad = \quad \frac{1}{f\,(D,\ D')}\phi(x,y)e^{ax+by} = \frac{e^{ax+by}}{f\,(D+a,\ D'+b)}\phi(x,y),$$

then proceed as type 3.

Type 5: If $F(x,\ y) = \sin ax \sin by$, then P.I is given by

$$\text{P.I} = \frac{1}{f\left(D^2,\ D'^2\right)}.\sin ax \sin by = \frac{\sin ax \sin by}{f\left(-a^2,\ -b^2\right)}, \qquad f\left(-a^2,\ -b^2\right) \neq 0.$$

Type 6: If $F(x,\ y) = \cos ax \cos by$, then P.I is given by

$$\text{P.I} = \frac{1}{f\left(D^2,\ D'^2\right)}.\cos ax \cos by = \frac{\cos ax \cos by}{f\left(-a^2,\ -b^2\right)}, \qquad f\left(-a^2,\ -b^2\right) \neq 0.$$

Note: If $f\left(-a^2,\ -b^2\right) = 0$ in both type, then proceed as the note given in Type 1.

General Method: If $f(D,\ D')z = F(x,\ y)$ is a given PDE, then the general method to find the P.I is given by

$$\text{P.I} = \frac{1}{(D - m_1 D')(D - m_2 D').....(D - m_n D')} \times F(x,\ y).$$

Then we shall use one of the following formulas

Formula 1:

$$\frac{1}{(D - mD')} \times F(x,\ y) = \int F(x,\ c - mx)dx, \quad \text{where} \quad c = y + mx.$$

Formula 2:

$$\frac{1}{(D + mD')} \times F(x,\ y) = \int F(x,\ c + mx)dx, \quad \text{where} \quad c = y - mx.$$

Depending up on the factor $D - mD'$ and $D + mD'$.

Example 1: Solve $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$.

Solution: The above equation can be written as

$$\left(D^3 - 3D^2D' + 4D'^3\right) z = e^{x+2y}, \qquad \text{here}$$

$$f\left(D,\ D'\right) = D^3 - 3D^2D' + 4D'^3 \quad \text{and from} \quad e^{x+2y}, \quad a = 1, \quad b = 2.$$

Put $D = m$ and $D' = 1$ in the above equation and write the auxiliary equation as

$$m^3 - 3m^2 + 4 = 0 \quad \Rightarrow (m+1)(m-2)^2 = 0 \quad \Rightarrow m = -1,\ 2,\ 2,$$

where 2 is repeated root. Therefore, the C.F. is

$$\boxed{C.F. = \phi_1(y-x) + x\phi_2(y+2x) + x^2\phi_2(y+2x).}$$

P.I.

$$\begin{aligned} P.I. &= \frac{1}{D^3 - 3D^2D' + 4D'^3} \times e^{x+2y}, \\ &\Rightarrow \frac{1}{(1)^3 - 3(1)^2(2) + 4(2)^3} \times e^{x+2y} \\ &= \frac{e^{x+2y}}{27}. \end{aligned}$$

$\therefore$ The complete solution is given by:

$$\boxed{C.F. + P.I. = \phi_1(y-x) + x\phi_2(y+2x) + x^2\phi_3(y+2x) + \frac{e^{x+2y}}{27}.}$$

Example 2: Solve $\left(D^3 - 7DD'^2 - 6D'^3\right) z = e^{2x+y}$.

Solution: The auxiliary equation is

$$m^3 - 7m - 6 = 0 \quad \Rightarrow (m+1)(m^2 - m - 6) = 0$$
$$\Rightarrow (m+1)(m-3)(m+2) = 0 \quad \Rightarrow m = -1,\ 3,\ -2$$

$\therefore$ The C.F. is

$$\boxed{C.F. = \phi_1(y-x) + \phi_2(y+3x) + \phi_3(y-2x).}$$

$$\begin{aligned} P.I. &= \frac{1}{D^3 - 7DD'^2 - 6D'^3} \times e^{2x+y} \Rightarrow \frac{1}{(2)^3 - 7(2)(1)^2 - 6(1)^3} \times e^{2x+y} \\ &= -\frac{e^{2x+y}}{12}. \end{aligned}$$

$$\boxed{z = C.F. + P.I. = \phi_1(y-x) + \phi_2(y+3x) + \phi_3(y-2x) - \frac{e^{2x+y}}{12}.}$$

Example 3: Solve $\left(D^2 - 4DD' + 4D'^2\right) z = e^{2x+y}$.

Solution: The auxiliary equation is

$m^2 - 4m + 4 = 0 \quad \Rightarrow (m-2)^2 \quad \Rightarrow m = 2,\ 2.$ $\therefore$ The C.F. is:

$$\boxed{C.F. = \phi_1(y+2x) + x\phi_2(y+2x).}$$

$$\begin{aligned} P.I. &= \frac{1}{D^2 - 4DD' + 4D'^2} \times e^{2x+y} \Rightarrow \frac{1}{(2)^2 - 4(2)(1) + 4(1)^2} \times e^{2x+y} \\ &= -\frac{e^{2x+y}}{0}. \end{aligned}$$

Hence we follow the Note.

Differentiate the denominator w.r.t D and multiply the numerator by x

$$P.I. = \frac{x}{2D - 4D'} \times e^{2x+y} = \frac{e^{2x+y}}{0}$$

$\therefore$ Once again differentiating the denominator w.r.t D and multiply the numerator by x, we get

$$P.I. = \frac{x^2}{2} \times e^{2x+y} \Rightarrow \frac{e^{2x+y}}{2}.$$

$\therefore$ The complete integral is given by:

$$\boxed{z = C.F. + P.I. = \phi_1(y + 2x) + x\phi_2(y + 2x) + \frac{e^{2x+y}}{2}.}$$

Example 4: Solve $\left(D^3 - 2D^2D'\right) z = \sin(x+2y) + 3x^2y$.

Solution: The auxiliary equation is
$m^3 - 2m^2 = 0 \quad \Rightarrow m^2(m-2) = 0 \quad \Rightarrow m = 0,\ 0,\ 2.$
$\therefore$ The C.F. is:

$$\boxed{C.F. = \phi_1(y) + x\phi_2(y) + \phi_2(y+2x).}$$

Here we have to find two P.I. corresponding to two terms $\sin(x+2y)$ and $3x^2y$ as in Type 2 and Type 3 respectively.

$$(P.I.)_1 = \frac{1}{D^3 - 2D^2D'} \times \sin(x+2y) \ \text{ and } \ (P.I.)_2 = \frac{1}{D^3 - 2D^2D'} \times 3x^2y$$

$$
\begin{aligned}
(P.I.)_1 &= \frac{1}{D^3 - 2D^2D'} \times \sin(x+2y) \\
&= \frac{1}{D.D^2 - 2D.DD'} \times \sin(x+2y) \\
&= \frac{1}{D. - (1)^2 - 2D.(-1.2)} \times \sin(x+2y) \\
&= \frac{1}{-D+4D} \times \sin(x+2y) = \frac{1}{3D} \times \sin(x+2y).
\end{aligned}
$$

As $\frac{1}{D}$ is integration w.r.t x, hence:

$$\boxed{(P.I.)_1 = -\tfrac{1}{3} \times \cos(x+2y).}$$

$$
\begin{aligned}
(P.I.)_2 &= \frac{1}{D^3 - 2D^2D'} \times 3x^2y = \frac{1}{D^3\left(1 - \frac{2D'}{D}\right)} \times 3x^2y \\
&= \frac{1}{D^3}\left(1 - \frac{2D'}{D}\right)^{-1} \times 3x^2y \quad \text{(Using Binomial expansion)} \\
&= \frac{1}{D^3}\left[1 + \frac{2D'}{D} - \frac{4D'^2}{D^2} + - - - - - - - - -\right] \times 3x^2y. \\
\Rightarrow \quad & \frac{1}{D^3}\left[3x^2y + \frac{6x^2}{D}\right] = \frac{1}{D^3}\left[3x^2y + 2x^3\right]
\end{aligned}
$$

Integrating w.r.t x thrice, we get $(P.I.)_2 = \frac{x^5y}{20} + \frac{x^6}{60}$. Finally we get

$$
\begin{aligned}
z &= C.F. + (P.I.)_1 + (P.I.)_2 \\
&= \phi_1(y) + x\phi_2(y) + \phi_2(y + 2x) - \frac{1}{3} \times \cos(x + 2y) + \frac{x^5y}{20} + \frac{x^6}{60}.
\end{aligned}
$$

Example 5: Solve $\left(D^2 - 2DD'\right) z = e^{2x} + x^3 y$.

Solution: The auxiliary equation is
$m^2 - 2m = 0 \quad \Rightarrow m(m-2) = 0 \quad \Rightarrow m = 0,\ 2.$
$\therefore$ The C.F. is:

$$\boxed{C.F. = \phi_1(y) + \phi_2(y + 2x).}$$

Here we have to find two P.I. corresponding to two terms e^{2x} and $x^3 y$ as in Type 2 and Type 3 respectively.

$$(P.I.)_1 = \frac{1}{D^2 - 2DD'} \times e^{2x} \quad \text{and} \quad (P.I.)_2 = \frac{1}{D^2 - 2DD'} \times x^3 y$$

$(P.I.)_1 = \frac{1}{D^2-2DD'} \times e^{2x} = \frac{1}{(2)^2-2(2)(0)} \times e^{2x} = \frac{e^{2x}}{4}.$

$$\begin{aligned}
(P.I.)_2 &= \frac{1}{D^2-2DD'} \times x^3y = \frac{1}{D^2\left(1-\frac{2D'}{D}\right)} \times x^3y \\
&= \frac{1}{D^2}\left(1-\frac{2D'}{D}\right)^{-1} \times x^3y \quad \text{(Using Binomial expansion)} \\
&= \frac{1}{D^2}\left[1+\frac{2D'}{D}-\frac{4D'^2}{D^2}+--------\right] \times x^3y. \\
\Rightarrow & \frac{1}{D^2}\left[x^3y+\frac{2x^3}{D}\right] = \frac{x^5y}{20}+\frac{x^6}{60}.
\end{aligned}$$

$$\boxed{z = C.F. + (P.I.)_1 + (P.I.)_2 = \phi_1(y) + \phi_2(y+2x) - \frac{e^{2x}}{4} + \frac{x^5y}{20} + \frac{x^6}{60}.}$$

Example 6: Solve $\left(D^2 + 3DD' + 2D'^2\right) z = x + y$.

Solution: The auxiliary equation is
$m^2 + 3m + 2 = 0 \quad \Rightarrow (m+1)(m+2) = 0 \quad \Rightarrow m = -1, \ -2.$
$\therefore$ The C.F. is:

$$\boxed{C.F. = \phi_1(y-x) + \phi_2(y-2x).}$$

$$\begin{aligned}(P.I.) &= \frac{1}{D^2 - 2DD'} \times (x+y) = \frac{1}{D^2\left(1 + \frac{3D'}{D} + \frac{D'^2}{D^2}\right)} \times (x+y) \\ &= \frac{1}{D^2}\left(1 + \frac{3D'}{D} + \frac{D'^2}{D^2}\right)^{-1} \times (x+y)\end{aligned}$$

$$= \frac{1}{D^2}\left[1+\frac{3D'}{D}+\frac{D'^2}{D^2}-(1+2.1.\frac{3D'}{D}.....)+.....\right]\times(x+y).$$

$$\Rightarrow \frac{1}{D^2}\left[\frac{9}{D}\right]=\frac{3x^3}{2}.$$

Hence P.I. is given by $\boxed{P.I.=\frac{3x^3}{2}.}$

$\therefore$ The complete integral is

$$\boxed{z=C.F.+P.I.=\phi_1(y-x)+\phi_2(y-2x)+\frac{3x^3}{2}.}$$

Example 6: Solve $\left(D^3 + D^2D' - DD'^2 - D'^3\right) z = e^x \cos 2y$.

Solution: The auxiliary equation is
$m^3 + m^2 - m - 1 = 0 \quad \Rightarrow (m+1)^2(m-1) = 0 \quad \Rightarrow m = -1, \ -1, \ 1.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y-x) + x\phi_2(y-x) + \phi_3(y+x).}$
In the expression e^x, we have $a = 1$ and $b = 0$, so

$$\begin{aligned}(P.I.) &= \frac{1}{D^3 + D^2D' - DD'^2 - D'^3} \times e^x \cos 2y \\ &= \frac{e^x}{(D+1)^3 + (D+1)^2D' - (D+1)D'^2 - D'^3} \times \cos 2y.\end{aligned}$$

$$= \quad e^x \frac{1}{(D+1)^3 + (D+1)^2 D' - (D+1)D'^2 - D'^3} \times \text{Real part of } e^{2iy}.$$

$$\Rightarrow \quad e^x \text{ Real part of} \frac{e^{2iy}}{1+2i+4+8i}$$

$$= \quad \frac{e^x}{5} \frac{1}{1+2i} \text{Real part of } (\cos 2y + i \sin 2y)$$

$$= \quad \frac{e^x}{5} \frac{1-2i}{1+4} \text{Real part of } (\cos 2y + i \sin 2y) = \frac{e^x}{25}(\cos 2y + 2\sin 2y)$$

$\therefore$ The complete integral is

$$\boxed{z = \phi_1(y-x) + x\phi_2(y-x) + \phi_3(y+x) + \tfrac{e^x}{25}(\cos 2y + 2\sin 2y).}$$

Example 6: Solve $\left(D^2 + DD' - 6D'^2\right) z = y \cos x$.

Solution: The auxiliary equation is
$m^2 + m - 6 = 0 \quad \Rightarrow (m-2)(m+3) = 0 \quad \Rightarrow m = 2, \ -3.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y+2x) + \phi_2(y-3x).}$

$$(P.I.) = \frac{1}{D^2 + DD' - 6D'^2} \times y \cos x = \frac{1}{(D-2D')(D+3D')} \times y \cos x$$

$$\Rightarrow \frac{1}{(D-2D')} \times \left[\int (c+3x) \cos x\right] dx \quad \text{where} \quad y = c + 3x$$

$$= \frac{1}{(D-2D')} \times [(c+3x)\sin x + 3\cos x] \quad \text{(integrating by parts)}$$

$$= \frac{1}{(D-2D')} \times [y \sin x + 3 \cos x].$$

Again

$$\begin{aligned}
& \frac{1}{(D-2D')} \times [y \sin x + 3 \cos x] \\
= & \int (c-2x) \sin x dx + 3 \int \cos x dx \quad \text{where} \quad y = c - 2x \\
= & \left[(c-2x)(-\cos x) - (-2) \int -\cos x dx + 3 \sin x \right] \\
= & -y \cos x - 2 \sin x + 3 \sin x = -y \cos x + \sin x.
\end{aligned}$$

$\therefore$ The complete solution is

$$\boxed{z = C.F. + P.I. = \phi_1(y+2x) + \phi_2(y-3x) - y \cos x + \sin x.}$$

Example 7: Solve $\left(D^2 - DD' - 2D'^2\right) z = (y-1)e^x$.

Solution: The auxiliary equation is
$m^2 - m - 2 = 0 \quad \Rightarrow (m+1)(m-2) = 0 \quad \Rightarrow m = -1,\ 2.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y-x) + \phi_2(y+2x).}$

$$\begin{aligned}
(P.I.) &= \frac{1}{D^2 - DD' - 2D'^2} \times (y-1)e^x \\
&= \frac{1}{(D+D')(D-2D')} \times (y-1)e^x \\
\Rightarrow &\ \frac{1}{(D+D')}\left[\frac{1}{(D-2D')}(y-1)e^x\right] \\
&= \frac{1}{(D+D')}\left[\int (c-2x-1)e^x\right] dx \quad \text{where } c = y+2x
\end{aligned}$$

$$= \frac{1}{(D+D')} \times [(c-2x-1)e^x + 2e^x] \quad \text{(integrating by parts)}$$
$$= \frac{1}{(D+D')} \times [(c-2x+1)e^x] = \frac{1}{(D+D')} \times [(y+1)e^x].$$

Again

$$\frac{1}{(D+D')} \times [(y+1)e^x] = \int (c+x+1)e^x dx \quad \text{where} \quad y = c+x$$
$$= (c+x+1)e^x - e^x \quad \text{integrating by part}$$
$$= (c+x)e^x = ye^x$$

$\therefore$ The complete solution is

$$\boxed{z = C.F. + P.I. = \phi_1(y-x) + \phi_2(y+2x) + ye^x.}$$

Example 8: Solve $\left(D^2 - DD'\right) z = \sin x \sin 2y$.

Solution: The auxiliary equation is
$m^2 - m = 0 \quad \Rightarrow m(m-1) = 0 \quad \Rightarrow m = 0,\ 1.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y) + \phi_2(y+x).}$

$$
\begin{aligned}
(P.I.) &= \frac{1}{D^2 - DD'} \times \sin x \sin 2y, \qquad \text{(As in Type 5, a=1, b=2)} \\
&= \frac{1}{D(D-D')} \times \sin x \sin 2y = \frac{D(D+D')}{D^2(D^2 - D'^2)} \times \sin x \sin 2y \\
\Rightarrow &\ \frac{D(D+D')}{-1^2\{-1^2 - (-2^2)\}} \times \sin x \sin 2y \\
&= \frac{D(D+D')}{-3} \times \sin x \sin 2y = \frac{(D^2 + DD')}{-3} \times \sin x \sin 2y
\end{aligned}
$$

As D and D' derivatives w.r.t x and y respectively, hence applying to $\sin x \sin 2y$, we get

$$\begin{aligned} \Rightarrow \quad & \frac{(D^2 + DD')}{-3} \times \sin x \sin 2y \\ = \quad & -\frac{1}{3}\left[-\sin x \sin 2y + 2\cos x \cos 2y\right]. \end{aligned}$$

$\therefore$ The complete solution is

$$\boxed{z = \phi_1(y) + \phi_2(y+x) - \tfrac{1}{3}\left[-\sin x \sin 2y + 2\cos x \cos 2y\right].}$$

Example 9: Solve $\left(D^2 - D'^2\right) z = \cos 2x \cos 3y$.

Solution: The auxiliary equation is
$m^2 - 4 = 0 \quad \Rightarrow (m-2)(m+2) = 0 \quad \Rightarrow m = 2, \ -2.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y+2x)+\phi_2(y-2x).}$

$$\begin{aligned}(P.I.) &= \frac{1}{D^2 - D'^2} \times \cos 2x \cos 3y, \qquad \text{(As in Type 5, a=1, b=2)} \\ &= \frac{\cos 2x \cos 3y}{-2^2 - 4(-3^2)} = \frac{\cos 2x \cos 3y}{32}.\end{aligned}$$

$\therefore$ The complete solution is

$$\boxed{z = \phi_1(y+2x) + \phi_2(y-2x) + \frac{\cos 2x \cos 3y}{32}.}$$

Example 10: Solve $\left(D^2 - 3DD' + 2D'^2\right) z = \sin x \cos y$.

Solution: The auxiliary equation is
$m^2 - 3m + 2 = 0 \quad \Rightarrow (m-1)(m-2) = 0 \quad \Rightarrow m = 1,\ 2.$
$\therefore$ The C.F. is: $\boxed{C.F. = \phi_1(y+x) + \phi_2(y+2x).}$

$$\begin{aligned}
(P.I.) &= \frac{\sin x \cos y}{D^2 - 3DD' + 2D'^2} = \frac{1}{2}\left[\frac{\sin(x+y) + \sin(x-y)}{D^2 - 3DD' + 2D'^2}\right] \\
&= \frac{1}{2}\left[\frac{\sin(x+y)}{D^2 - 3DD' + 2D'^2} + \frac{\sin(x-y)}{D^2 - 3DD' + 2D'^2}\right] \\
&\qquad (P.I.)_1 \qquad + \qquad (P.I.)_2
\end{aligned}$$

As in Type 2, we can have

$$
\begin{aligned}
(P.I.)_1 &= \frac{1}{2}\frac{\sin(x+y)}{D^2-3DD'+2D'^2} = \frac{1}{2}\frac{\sin(x+y)}{-1^2-3(-1.1)+2(-1^2)} \\
&= \frac{1}{2}\frac{\sin(x+y)}{0} \Rightarrow \frac{1}{2}\frac{\sin(x+y)}{(D-2DD')(D-D')} \\
&= \frac{1}{2}\frac{1}{(D-2DD')}\left[\frac{1}{(D-D')}\sin(x+y)\right] \\
&= \frac{1}{2}\frac{1}{(D-2DD')}\left[\int \sin(x+c-x)\right]dx \quad \text{where} \quad c=y+x \\
&= \frac{1}{2}x\sin c = \frac{x\sin(x+y)}{2}
\end{aligned}
$$

As in Type 2, we can have

$$\begin{aligned}(P.I.)_2 &= \frac{1}{2}\frac{\sin(x-y)}{D^2-3DD'+2D'^2}\\ &= \frac{1}{2}\frac{\sin(x-y)}{-1-3-2}\\ &= \frac{1}{2}\frac{\sin(x-y)}{-6}\\ &= \frac{\sin(x-y)}{-12}.\end{aligned}$$

$\therefore$ The complete solution is $C.F. + (P.I.)_1 + (P.I.)_2$

$$\boxed{z = \phi_1(y+x) + \phi_2(y+2x) + \frac{x\sin(x+y)}{2} - frac\sin(x-y)12.}$$